

AEGIS // Autonomous Multi-Agent Intelligence System

System Capability Audit, Architecture Overview & Improvement Roadmap

1. AEGIS Kya Hai? (Executive Summary)

AEGIS ek next-generation **Autonomous Enterprise Intelligence & Multi-Agent Orchestration Platform** hai. Yeh business workflows, enterprise data, aur daily operational decision-making ko AI agents ki network mesh dwara handle karne ke liye banaya gaya hai.

2. AEGIS me Kya-Kya Features Working Hain?

Component / Module	Status & Key Capabilities
27-Agent Orchestrator Mesh	Multi-LLM routing (Groq Llama 3.3, NVIDIA NIM, HuggingFace). Prompts ke intent ko auto-classify karke specialized agents (Coder, DevOps, Architect, Sales, etc.) ko tasks automatically delegate karta hai.
Workspace RAG Chat	Live streaming WebSocket chat (<code>ws://localhost:8000/api/v1/chat/stream</code>). High-speed vector search powered by Qdrant DB for citing sources in answers.
Document Ingestion Pipeline	PDF/TXT document upload, chunking, and embedding creation stored directly into PostgreSQL (Neon DB) & Qdrant.
Real-Time Metal Rates Integration	Real-time Gold & Silver live rate tracking from external financial APIs with automatic currency conversion (INR).
Role-Based Access Control (RBAC)	Customizable access profiles for Founder, Sales Manager, Marketing, Finance, and Support tiers.

3. Abhi Kya-Kya Lackings & Mock Data Parts Hain?

- Stripe & HubSpot BI Feeds:** Dashboard yield metrics fallback mock JSON data par chal rahe hain jab tak environment keys (Stripe API / HubSpot Key) production mode me live attach nahi ki jaati.
- Agent Sequential Workflows:** Multi-agent pipeline concurrent execution generate karti hai, par complex long-running persistent memory graphs (like LangGraph execution traces) abhi in-memory handled hain.
- Live Database Auth:** User login and onboarding local app state me persistent authentication cookies ke bajaye standard JWT session token fallback run kar rahi hai.
- Customer Support Automation:** Support tickets board resolution draft toh generate kar leta hai, par live Zendesk/Intercom integration pending hai.

4. AEGIS me Kya-Kya Improvements Ki Ja Sakti Hain?

Priority Area	Proposed Enhancements & Action Plan
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1. Production BI Connectors	Stripe Webhooks & HubSpot OAuth sync set up karke live revenue analytics aur lead pipelines connect karna.
2. Persistent Agent Memory Graph	LangGraph / Redis Checkpointers introduce karke persistent conversation states aur multi-turn agent execution memory store karna.
3. Automated Action Execution	Agents ko sirf code/text response dene tak simit na rakh kar actual API execution (e.g. automatic Email dispatch, GitHub Commit push) capabilities dena.
4. Advanced Multi-Modal Ingestion	Excel (XLSX), CSV financial ledgers, audio meeting transcripts, aur images ki automatic vision-parsing RAG pipeline start karna.